

PES UNIVERSITY, BANGALORE
Department of Computer Science and Engineering

Jackfruit Phase-1

Software Requirements Specification

for

Role-Specific AI Voice Assistant Framework for Organizations and Entities

Version 1.0 approved

Team details

Team Name	Team ID	Team Member 1 (Name & SRN)	Team Member 2 (Name & SRN)	Team Member 3 (Name & SRN)	Team Member 4 (Name & SRN)
NexGen	43	Varun Kumar H L PES2UG24AM231	Tejaswi Ganapati Hegde PES2UG24AM171	Yashas V PES2UG24AM188	Sri Kanth G PES2UG24AM161

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Prepared by:

Varun Kumar H L
Tejaswi Ganapati Hegde
Yashas V
Sri Kanth G

Table of Contents

Table of Contents	ii
Revision History	ii
1. Introduction	1
1.1 Purpose	1
1.2 Intended Audience and Reading Suggestions	1
1.3 Product Scope	1
1.4 References	1
2. Overall Description	2
2.1 Product Perspective	2
2.2 Product Functions	2
2.3 User Classes and Characteristics	2
2.4 Operating Environment	2
2.5 Design and Implementation Constraints	2
2.6 Assumptions and Dependencies	3
3. External Interface Requirements	3
3.1 User Interfaces	3
3.2 Software Interfaces	3
3.3 Communications Interfaces	3
4. Analysis Models	4
5. System Features	4
5.1 – 5.13 (see below)	4
6. Other Nonfunctional Requirements	10
6.1 Performance Requirements	10
6.2 Safety Requirements	10
6.3 Security Requirements	10
6.4 Software Quality Attributes	11
6.5 Business Rules	11
7. Other Requirements	12
Appendix A: Glossary	12
Appendix B: Field Layouts	13
Appendix C: Requirement Traceability Matrix	13

Revision History

Name	Date	Reason For Changes	Version
NexGen	01-09-2026	Initial draft of SRS	1.0

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the **Role-Specific AI Voice Assistant Framework for Organizations and Entities** (Project ID 42), being developed as part of the Software Engineering (UE24CS341A) mini-project (Jackfruit) at PES University. The system is a customizable, voice-enabled virtual assistant framework that can be configured and deployed to serve multiple functional roles within an organization – such as an HR Assistant, an IT Support Assistant, and an Admissions Assistant – from a single underlying platform. Each role assistant is tailored through a configurable role profile that defines its supported intents, conversation flows, and the internal organizational tools or systems it is authorized to interact with.

This document covers Version 1.0 of the proposed system and is intended to guide the requirements, planning, design, implementation, and testing phases of the project as described in the Jackfruit mini-project guidelines.

1.2 Intended Audience and Reading Suggestions

- **Developers:** use this document to understand the functional requirements, system behavior, interfaces, and operating environment needed to implement the framework.
- **Project Manager:** uses this document to understand scope, priorities, and dependencies for sprint and iteration planning under the Agile methodology mandated for the course.
- **Testers:** use the functional and non-functional requirements in Sections 5 and 6 to derive unit, integration, and system test cases (Jackfruit Phase-3).
- **End Users (Employees):** can refer to this document to understand the voice-assistant interactions available to them for each configured role.
- **Organization Administrators:** use this document to understand role-configuration, integration, and monitoring capabilities.
- **Project Team and Evaluators:** use this document to review scope, analysis models, system features, and quality requirements as part of the Jackfruit evaluation process.

Readers are advised to review Section 2 (Overall Description) first for context, then Section 5 (System Features) for the detailed functional requirements relevant to their role assistant of interest, followed by Section 6 for cross-cutting non-functional requirements.

1.3 Product Scope

The **Role-Specific AI Voice Assistant Framework** is a web-based platform that allows an organization to deploy one AI voice/chat assistant per functional role (for example HR, IT Support, and Admissions) without building a separate assistant from scratch for each department. A single underlying conversational engine is reused across roles; only the intents, sample utterances, workflow mappings, and integration endpoints differ per role profile.

The system enables employees (users) to:

- Register and log in to the system.
- Select the role-specific assistant they wish to interact with (e.g., HR Assistant, IT Support Assistant, Admissions Assistant).
- Interact with the assistant using voice or typed natural-language queries.
- Have their query converted to text, its intent recognized, and a corresponding workflow executed.
- Receive a spoken and/or textual response generated by the assistant.
- Escalate a query to a human agent when the assistant cannot resolve it.

- View their past conversation history.

For organization administrators, the system enables them to:

- Create and configure role profiles (intents, sample utterances, and workflow mappings) for new or existing organizational roles.
- Configure and manage integrations with internal organizational systems (e.g., HRMS, IT ticketing/ITSM, admissions/CRM systems) through configurable API connectors.
- Manage role-based access control for users, administrators, and human agents.

Monitor assistant usage and view analytics and reports through an administrative dashboard.

Review escalated conversations and reassign them to available human agents.

The primary objective of the system is to reduce the effort and cost of building and maintaining separate AI assistants for each organizational function by providing a single, configurable, role-aware voice assistant framework, thereby improving employee self-service, reducing repetitive query load on human staff, and enabling faster resolution of common role-specific requests.

1.4 References

- [1] Team, Role-Specific AI Voice Assistant Framework – Use Case and Activity Diagrams, Version 1.0, 2026 (attached in Section 4 and in the project GitHub repository).
- [2] Jackfruit Phase-1 Software Requirements Specification Template, Version 1.0, provided by the Department of Computer Science and Engineering, PES University, August 2026.
- [3] Jackfruit Mini-Project Guidelines, Subject Code UE24CS341A, Department of Computer Science and Engineering, PES University, 2026.
- [4] Team, Role-Specific AI Voice Assistant Framework – Functional and Non-Functional Requirement List, Version 1.0, 2026 (attached in project GitHub repository).

2. Overall Description

2.1 Product Perspective

The Role-Specific AI Voice Assistant Framework is a new, self-contained academic prototype developed as part of this project. It is not a replacement for, or extension of, any existing production assistant, but is designed to demonstrate how a single conversational-AI backend can be parameterized to serve multiple organizational roles. The system follows a client-server, layered architecture in which users interact with the assistant through a web-based voice/chat interface; a backend server processes speech, recognizes intent, executes the mapped workflow, and returns a response, while a database persists role profiles, users, conversations, and integration configuration.

The system consists of three primary user roles:

- **User (Employee):** Logs in, selects a role assistant, converses with it by voice or text, receives responses, views conversation history, and escalates unresolved queries.
- **Organization Administrator:** Configures role profiles, manages integrations with internal systems, manages role-based access control, and monitors usage through an administrative dashboard.
- **Human Agent:** Receives and handles conversations escalated by the assistant when it cannot resolve a query with sufficient confidence.

Figure 2.1 below illustrates the high-level architecture of the system.

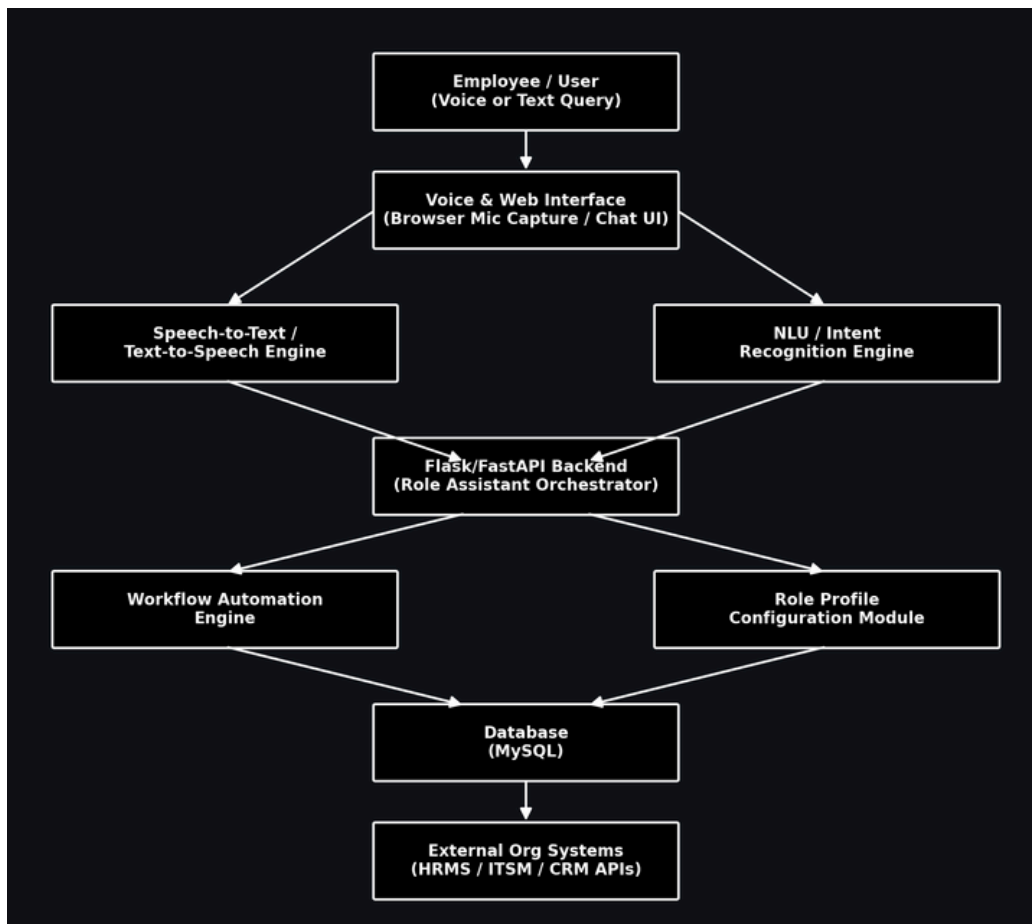


Figure 2.1: High-level architecture of the Role-Specific AI Voice Assistant Framework.

The system will be implemented using Python as the primary programming language, with Flask (or FastAPI) as the backend web framework, a speech-to-text/text-to-speech pipeline for voice interaction, and an

NLU/intent-recognition component for understanding user queries.

The Role-Specific AI Voice Assistant Framework provides the following major functions:

2.2 Product Functions

User Management

- User registration and login.
- Role-based access control for users, administrators, and human agents.

Role Assistant Configuration

- Creation and configuration of role profiles (e.g., HR, IT Support, Admissions) with associated intents, sample utterances, and workflow mappings.
- Selection of a role assistant by the user before starting a conversation.

Conversational Interaction

- Voice input capture and speech-to-text conversion.
- Text-based query input as an alternative to voice.
- Intent recognition and entity extraction from the user query.
- Text-to-speech conversion of the generated response.

Workflow Automation

- Mapping of recognized intents to organization-specific workflows or API actions.
- Execution of the mapped workflow, including calls to configured internal-tool integrations.

Integration Management

- Configuration and management of connectors to internal organizational systems (HRMS, ITSM, CRM, etc.).

Escalation Management

- Escalation of low-confidence or unresolved queries to a human agent.
- Handling of escalated conversations by human agents.

Conversation History and Notifications

- Storage and retrieval of conversation history.
- Notifications to users regarding escalation and resolution updates.

Administrative Management

- Administrative dashboard for monitoring assistant usage.
- Reports and analytics on intents handled, escalation rate, and role-wise usage.
- Search and filtering of conversations and users.

These functions collectively support the end-to-end lifecycle of a role-specific conversation: role selection, query capture, intent recognition, workflow execution or escalation, response generation, and logging.

2.3 User Classes and Characteristics

i) User (Employee)

Users are employees or members of the organization who interact with a role-specific assistant to get quick answers or complete simple transactional requests (e.g., check leave balance, raise an IT ticket, ask an admissions query). Users are expected to have basic computer/smartphone literacy and require no specialized

technical expertise.

- Register and log in.
- Select a role assistant.
- Interact using voice or text.
- View responses and conversation history.
- Escalate a query to a human agent.

ii) Organization Administrator

Administrators are authorized personnel responsible for configuring role assistants and managing the framework for their organization. They are expected to have moderate technical familiarity with the organization's internal systems (HRMS, ITSM, CRM) in order to configure integrations.

- Create and configure role profiles, intents, and workflow mappings.
- Configure and manage integrations with internal systems.
- Manage role-based access control.
- Access the administrative dashboard and reports.

iii) Human Agent

Human agents are support staff who receive conversations escalated by the assistant. They require access only to the escalation queue and the relevant conversation context.

- View and accept escalated conversations.
- Respond to and resolve escalated queries.

Role-based access control is used to ensure that each user class can access only the functions permitted for its respective role.

2.4 Operating Environment

Client Environment

- Desktop, laptop, or smartphone capable of running a modern web browser with microphone access.
- Modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge.
- Stable internet/network connectivity for voice streaming and API calls.

Application Environment

- Python with Flask (or FastAPI) as the backend web application framework.

Server Environment

- A server capable of hosting the Python runtime, the web framework, and the speech-processing components.

Database Environment

MySQL will be used as the database management system to store user information, role profiles, intents, workflow mappings, integration configuration, conversation history, escalation records, and analytics data.

Development Environment

Development and testing will be performed on standard computer systems supporting Python, Flask, MySQL, a speech-to-text/text-to-speech library or API, and the required web-server components.

2.5 Design and Implementation Constraints

- **Technology Stack Constraint:** The system shall be implemented using Python as the primary programming language, Flask/FastAPI as the web application framework, and MySQL as the database management system.
- **Speech Processing Constraint:** Speech-to-text and text-to-speech functionality shall use an available open-source library or a free-tier cloud speech API, given the academic scope of the project.
- **Time and Resource Constraint:** The project shall be developed and completed within the stipulated academic timeline (Jackfruit Phases 1–3) and the available team resources.
- **Prototype Scope Constraint:** The prototype shall demonstrate a limited number of role assistants (HR Assistant, IT Support Assistant, and Admissions Assistant) rather than an exhaustive set of organizational roles.
- **External Integration Constraint:** Integrations with internal organizational systems (HRMS, ITSM, CRM) shall be simulated using mock services or sandbox/test APIs, since the project shall not depend on access to a real production organizational system.
- **Data Privacy Constraint:** User personal information and conversation content shall be stored securely and shall not be unnecessarily exposed through publicly accessible interfaces.
- **Browser Compatibility Constraint:** The web application shall support commonly used modern web browsers with microphone-access permission support, including Google Chrome, Mozilla Firefox, and Microsoft Edge.
- **Scalability Constraint:** The system is designed as an academic proof-of-concept and is not intended to support large-scale, multi-tenant, enterprise-wide deployment as part of the current project.

2.6 Assumptions and Dependencies

Assumptions:

- Users are assumed to have access to a device with a supported modern web browser, a working microphone (for voice interaction), and a working internet connection.
- Users are assumed to provide valid information during registration and login and to maintain the confidentiality of their account credentials.
- Organization administrators are assumed to have sufficient knowledge of their organization's processes to correctly configure role profiles, intents, and workflow mappings.
- The predefined intents configured for each role are assumed to cover the majority of queries handled by that role assistant; unmatched queries are handled through escalation.
- Human agents are assumed to be available, at least during testing and demonstration, to receive escalated conversations.

Dependencies:

- The system depends on the availability of the Flask/FastAPI framework and the Python runtime environment for application execution.
- The system depends on the availability and proper functioning of the MySQL database for storing and retrieving user, role-profile, conversation, and integration data.
- The voice-interaction functionality depends on the availability of a speech-to-text and text-to-speech library or service.
- The workflow-execution functionality depends on the availability of the configured internal-system integrations (mocked or sandboxed for this project).
- The system depends on a stable network connection for communication between the client, the backend, and the database server.

3. External Interface Requirements

3.1 User Interfaces

The Role-Specific AI Voice Assistant Framework shall provide a web-based user interface accessible through supported modern web browsers. Separate interface views shall be provided for users, administrators, and human agents according to their authorized functions.

User Interface:

- **Registration and Login:** Forms for creating an account and securely logging in.
- **Role Selection Screen:** Displays the list of role assistants (e.g., HR Assistant, IT Support Assistant, Admissions Assistant) available to the user.
- **Conversation Interface:** A chat-style interface with a microphone control for voice input, a text box for typed queries, and a transcript of the ongoing conversation with spoken playback of responses.
- **Conversation History Interface:** Displays previously held conversations and their outcomes.
- **Escalation Notification:** Displays a message when a query is escalated to a human agent and notifies the user when the agent responds.

Administrator Interface:

- **Administrator Login:** Provides secure access to administrative functions.
- **Role Profile Configuration Interface:** Allows creation and editing of role profiles, intents, sample utterances, and workflow mappings.
- **Integration Management Interface:** Allows configuration of connectors to internal organizational systems.
- **Access Control Interface:** Allows management of user, administrator, and human-agent roles and permissions.
- **Admin Dashboard:** Provides an overview of assistant usage and key statistics.
- **Reports and Analytics Interface:** Provides analytical reports on intents, escalation rates, and role-wise usage.

Human Agent Interface:

- **Escalation Queue Interface:** Lists conversations escalated to human agents.
- **Conversation Handling Interface:** Allows the agent to view conversation context and respond to the user.

General UI Guidelines:

- The interfaces shall use a simple, clear, and consistent layout across all screens.
- Navigation elements shall be placed consistently to allow users to move easily between the dashboard, role selection, conversation, history, and logout functions.
- Standard controls such as Submit, Cancel, Speak, Send, Search, Filter, and Logout shall be used where applicable.
- The system shall perform input validation and display clear, understandable error messages for invalid, incomplete, or unrecognized input.
- The interface shall provide clear visual indication of microphone status (listening/idle) and of the current conversation state.
- The interface shall provide accessibility considerations, including readable text, clear labels, sufficient contrast, and keyboard-accessible controls, in addition to voice interaction.
- Access to screens and functions shall be controlled according to the user's authorized role.

3.2 Software Interfaces

The Role-Specific AI Voice Assistant Framework shall interact with the following software components:

Backend Framework:

Flask (or FastAPI) shall be used as the web application framework for implementing the server-side logic, handling requests from the client interface, orchestrating speech processing and intent recognition, executing workflows, and communicating with the database.

Database:

MySQL shall be used for persistent storage of user details, role profiles, intents, workflow mappings, conversation history, escalation records, integration configuration, and analytics data.

Speech-to-Text / Text-to-Speech Engine:

The system shall use a speech-to-text component to transcribe spoken user queries into text and a text-to-speech component to convert generated textual responses into spoken audio played back to the user.

NLU / Intent Recognition Engine:

The system shall use a natural-language-understanding component to classify the recognized intent of a user query and extract relevant entities, using the intents configured for the currently selected role.

Internal Organizational System Integrations:

The workflow automation engine shall communicate with configurable internal-tool connectors (for example, an HRMS API for leave balance queries, an ITSM API for raising support tickets, or a CRM/admissions API for inquiry status) using REST-based API calls. For this academic prototype, these integrations shall be represented using mock or sandbox services.

Authentication and Authorization:

The system shall provide authentication for registration and login, and role-based access control shall access to administrative and human-agent functions.

3.3 Communications Interfaces

restrict

- **Client–Server Communication:** Communication between the client browser and the Flask/FastAPI backend shall use HTTP/HTTPS, with HTTPS preferred for deployment to protect data in transit.
- **Voice Streaming:** Audio captured through the browser microphone shall be transmitted to the speech-to-text component using standard web audio APIs and HTTP(S)/WebSocket communication as appropriate.
- **Data Transfer:** User, role-profile, conversation, and integration data shall be exchanged between the client, backend, and database using standard web request/response mechanisms.
- **Notification Communication:** The system shall notify users regarding escalation and resolution events through the web interface.
- **Network Requirement:** A stable internet connection shall be required to access the application and to stream voice input/output.
- **Communication Security:** Communication containing user credentials, conversation content, and integration data shall be protected using secure communication mechanisms such as HTTPS during deployment.

(Hardware interfaces will not be implemented for this project.)

4. Analysis Models

Use Case Diagram

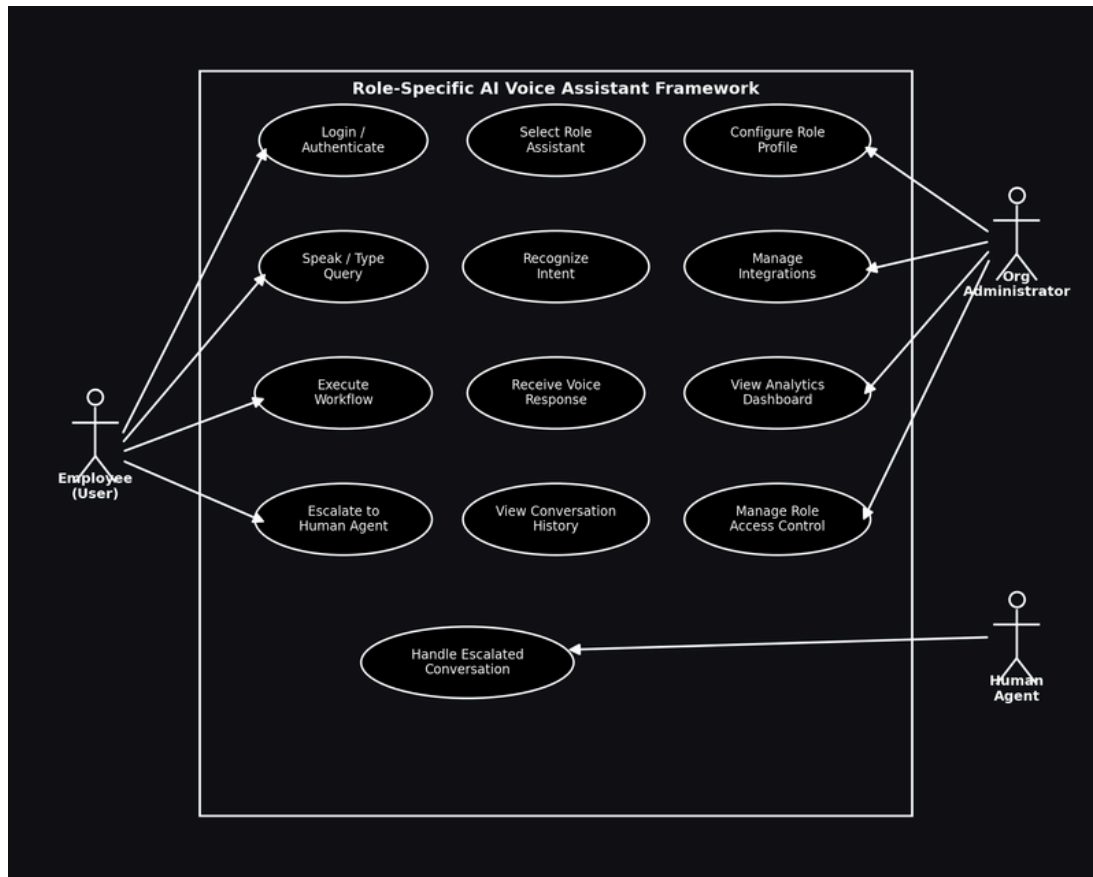


Figure 4.1: Use case diagram of the Role-Specific AI Voice Assistant Framework.

The use case diagram illustrates the interactions between the three primary actors – Employee (User), Organization Administrator, and Human Agent – and the major functions provided by the system. Employees primarily interact with role selection, query submission, workflow execution, response delivery, and escalation. Administrators configure role profiles, manage integrations and access control, and monitor usage through the analytics dashboard. Human agents handle escalated conversations.

Activity Diagram

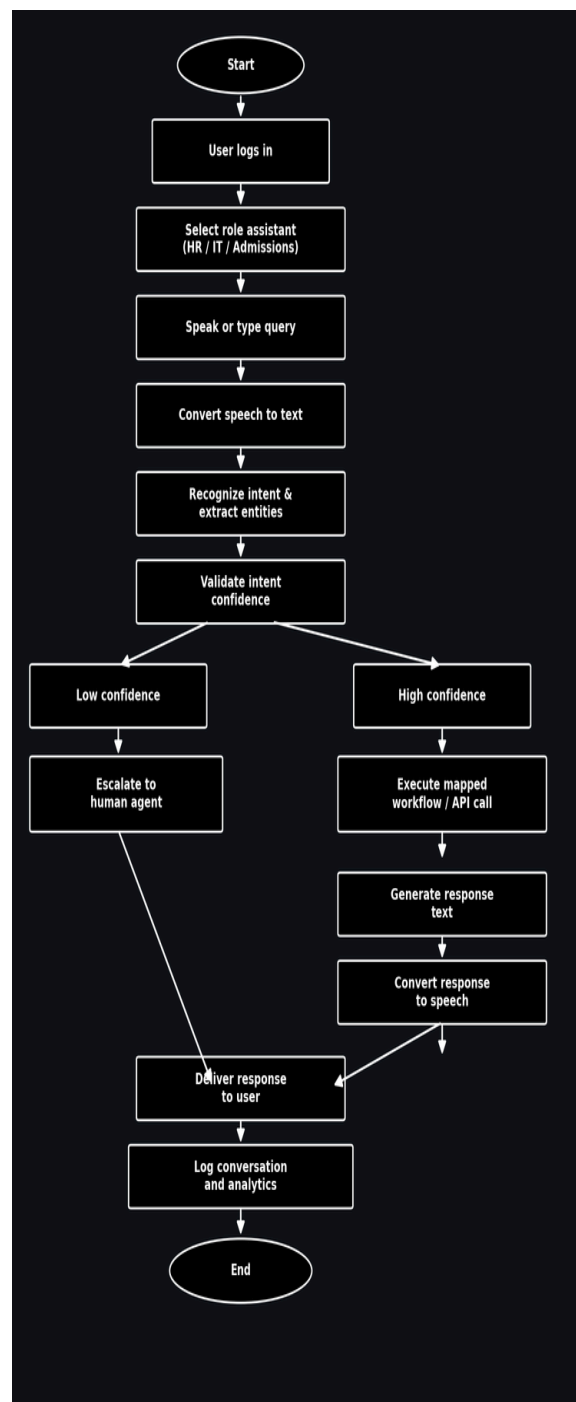


Figure 4.2: Activity diagram of a role-specific conversation.

The activity diagram represents the workflow of a single conversation with the assistant, beginning with user login and role selection, followed by capturing the spoken or typed query, converting speech to text, recognizing intent, and validating confidence. High-confidence intents are routed to the workflow automation engine for execution and a spoken response; low-confidence intents are escalated to a human agent. The conversation concludes with logging of the interaction for analytics.

5. System Features

5.1 User Registration and Login

5.1.1 Description and Priority

This feature allows employees, administrators, and human agents to create an account and securely log in to the framework. Authentication is required to access protected functions such as role assistant selection, conversation history, and administrative or agent functions.

Priority: High Benefit: 9, Penalty: 9, Cost: 4, Risk: 2

5.1.2 Stimulus/Response Sequences

- User selects "Register" → System displays the registration form.
- User enters the required registration details and submits the form → System validates the information.
- Registration information is valid → System creates the account and displays a confirmation.
- User enters valid login credentials and selects "Login" → System authenticates the user and redirects to the appropriate dashboard based on role.
- User enters invalid login credentials → System rejects the login attempt and displays an appropriate error message.
- An unauthenticated user attempts to access a protected function → System denies access and prompts the user to log in.

5.1.3 Functional Requirements

REQ-1.1	The system shall allow a new user to register by providing the required registration information.
REQ-1.2	The system shall validate the registration information before creating an account.
REQ-1.3	The system shall prevent registration using information already associated with an existing account and shall display an appropriate error message.
REQ-1.4	The system shall allow a registered user to log in using valid authentication credentials.
REQ-1.5	The system shall deny access when invalid login credentials are provided and shall display an appropriate error message.
REQ-1.6	
REQ-1.7	The system shall restrict access to protected system functions to authenticated users.
	The system shall provide access to system functions based on the authenticated user's authorized role (User, Administrator, or Human Agent).

5.2 Role Assistant Selection

5.2.1 Description and Priority

This feature allows a logged-in user to select which role-specific assistant (e.g., HR Assistant, IT Support Assistant, Admissions Assistant) they wish to converse with, based on the role profiles configured for the organization.

Priority: High Benefit: 8, Penalty: 6, Cost: 3, Risk: 2

5.2.2 Stimulus/Response Sequences

- User logs in → System displays the list of available role assistants configured for the organization.

- User selects a role assistant → System loads the intents and workflow mappings associated with that role and opens the conversation interface.
- No role assistants are configured → System displays an appropriate message indicating that no assistants are currently available.

5.2.3 Functional Requirements

REQ-2.1	The system shall display the list of role assistants currently configured and active for the organization.
REQ-2.2	The system shall allow a user to select a role assistant before starting a conversation.
REQ-2.3	The system shall load the intents, sample utterances, and workflow mappings associated with the selected role profile.
REQ-2.4	The system shall display an appropriate message when no role assistants are available for selection.

5.3 Voice and Text Query Capture

5.3.1 Description and Priority

This feature allows a user to submit a query to the selected role assistant either by speaking through the device microphone or by typing text, and converts spoken input into text for downstream processing.

Priority: High Benefit: 9, Penalty: 8, Cost: 6, Risk: 5

5.3.2 Stimulus/Response Sequences

- User selects the microphone control and speaks a query → System captures the audio and converts it to text using the speech-to-text engine.
- User types a query into the text box and submits it → System accepts the typed text directly.
- The captured audio is unclear or empty → System displays an appropriate message asking the user to repeat the query.
- Speech-to-text conversion fails due to a technical error → System displays an error message and allows the user to retry or type the query instead.

5.3.3 Functional Requirements

REQ-3.1	The system shall allow a user to submit a query using voice input captured through the device microphone.
REQ-3.2	The system shall allow a user to submit a query by typing text as an alternative to voice input.
REQ-3.3	The system shall convert captured voice input into text using a speech-to-text engine.
REQ-3.4	The system shall display an appropriate message when voice input cannot be captured or converted successfully.
REQ-3.5	The system shall pass the resulting query text to the intent recognition engine for further processing.

5.4 Intent Recognition and Entity Extraction

5.4.1 Description and Priority

This feature analyzes the text of a user query, using the intents configured for the currently selected role, to determine the user's intent and extract relevant entities (such as dates, ticket categories, or employee identifiers) needed to execute the corresponding workflow.

Priority: High Benefit: 9, Penalty: 8, Cost: 7, Risk: 6

5.4.2 Stimulus/Response Sequences

- A query is received in text form → System analyzes the text against the intents configured for the selected role assistant.
- An intent is recognized with sufficient confidence → System extracts relevant entities from the query and passes the intent and entities to the workflow automation engine.
- No intent is recognized with sufficient confidence → System flags the query for escalation to a human agent. Required entities are missing from the query → System prompts the user for the missing information before proceeding.

5.4.3 Functional Requirements

REQ-4.1	The system shall analyze the text of each user query to determine its intent, using the intents configured for the currently selected role.
REQ-4.2	The system shall extract relevant entities from the user query required to execute the corresponding workflow.
REQ-4.3	The system shall compute a confidence score for the recognized intent.
REQ-4.4	The system shall flag a query for escalation when the intent confidence score falls below a configured threshold.
REQ-4.5	The system shall prompt the user for missing required entities before executing the mapped workflow.

5.5 Workflow Automation and Execution

5.5.1 Description and Priority

This feature executes the workflow mapped to a recognized intent, which may include invoking an internal organizational system integration (e.g., checking leave balance in an HRMS, raising a ticket in an ITSM) and preparing the information needed to generate a response.

Priority: High Benefit: 9, Penalty: 8, Cost: 7, Risk: 6

5.5.2 Stimulus/Response Sequences

- An intent and its extracted entities are received → System looks up the workflow mapped to that intent for the current role.
- The mapped workflow requires an internal-tool integration call → System invokes the configured connector with the relevant entities.
- The integration call succeeds → System receives the result and prepares it for response generation.
- The integration call fails or times out → System logs the failure and prepares a fallback response, optionally escalating the query to a human agent.

5.5.3 Functional Requirements

REQ-5.1	The system shall map each recognized intent to a corresponding workflow definition for the current role.
REQ-5.2	The system shall execute the mapped workflow, including invoking configured internal-tool integrations where required.
REQ-5.3	The system shall pass the extracted entities as parameters to the invoked integration.
REQ-5.4	The system shall handle integration failures gracefully and shall log the failure for administrative review.
REQ-5.5	The system shall escalate the query to a human agent when the mapped workflow cannot be completed successfully.

5.6 Response Generation and Text-to-Speech

5.6.1 Description and Priority

This feature generates a natural-language response based on the outcome of the executed workflow and converts it into spoken audio, in addition to displaying it as text in the conversation interface.

Priority: High Benefit: 8, Penalty: 6, Cost: 5, Risk: 4

5.6.2 Stimulus/Response Sequences

- A workflow completes execution → System generates a natural-language response describing the outcome.
- The generated response is ready → System converts the response text into speech using the text-to-speech engine and plays it back to the user.
- Text-to-speech conversion fails → System displays the response as text only and logs the failure.

5.6.3 Functional Requirements

REQ-6.1	The system shall generate a natural-language response based on the result of the executed workflow.
REQ-6.2	The system shall convert the generated response into speech using a text-to-speech engine.
REQ-6.3	The system shall display the generated response as text in the conversation interface in addition to playing the spoken response.
REQ-6.4	The system shall display the response as text only when text-to-speech conversion is unavailable or fails.

5.7 Role Profile Configuration

5.7.1 Description and Priority

This feature allows an organization administrator to create and maintain role profiles, each consisting of a role name, a set of intents with sample utterances, and the workflow mappings associated with each intent.

Priority: High Benefit: 9, Penalty: 7, Cost: 7, Risk: 6

5.7.2 Stimulus/Response Sequences

- Administrator selects "Create Role Profile" → System displays a form to enter the role name and description.
- Administrator adds an intent with sample utterances and a workflow mapping → System validates and stores the intent definition under the role profile.

- Administrator submits the completed role profile → System validates the configuration and activates the role assistant for user selection.
- Administrator edits an existing role profile → System updates the stored configuration and reflects the change for future conversations.
- Required configuration fields are missing or invalid → System displays an appropriate error message.

5.7.3 Functional Requirements

REQ-7.1	The system shall allow an authorized administrator to create a new role profile with a name and description.
REQ-7.2	The system shall allow an administrator to define one or more intents, with sample utterances, under a role profile.
REQ-7.3	The system shall allow an administrator to map each intent to a corresponding workflow.
REQ-7.4	The system shall validate role-profile configuration before activating a role assistant.
REQ-7.5	The system shall allow an administrator to edit or deactivate an existing role profile.
REQ-7.6	The system shall make an activated role profile available for selection by users.

5.8 Integration Management

5.8.1 Description and Priority

This feature allows an organization administrator to configure and manage connectors to internal organizational systems (such as HRMS, ITSM, or CRM/admissions systems) that role-assistant workflows can invoke.

Priority: Medium Benefit: 7, Penalty: 5, Cost: 7, Risk: 6

5.8.2 Stimulus/Response Sequences

- Administrator selects "Manage Integrations" → System displays the list of currently configured integrations.
- Administrator adds a new integration by providing connection details (endpoint, authentication, supported operations) → System validates and stores the integration configuration.
- Administrator tests an integration → System attempts a sample call and displays the result.
- Administrator removes or disables an integration → System prevents workflows from invoking that integration going forward.

5.8.3 Functional Requirements

REQ-8.1	The system shall allow an authorized administrator to configure a connector to an internal organizational system.
REQ-8.2	The system shall validate integration configuration details before activation.
REQ-8.3	The system shall allow an administrator to test a configured integration and view the result.
REQ-8.4	The system shall allow an administrator to disable or remove an existing integration.
REQ-8.5	The system shall prevent workflows from invoking an integration that has been disabled or removed.

5.9 Escalation to Human Agent

5.9.1 Description and Priority

This feature routes a query to a human agent when the assistant cannot recognize the intent with sufficient confidence or cannot successfully complete the mapped workflow, ensuring the user's request is still addressed.

Priority: High Benefit: 8, Penalty: 7, Cost: 5, Risk: 4

5.9.2 Stimulus/Response Sequences

- A query is flagged for escalation → System adds the conversation to the escalation queue and notifies available human agents.
- A human agent accepts an escalated conversation → System assigns the conversation to that agent and displays the relevant context.
- The human agent responds to the user → System delivers the agent's response to the user through the conversation interface.
- No human agent is available → System notifies the user of the delay and keeps the conversation in the queue.

5.9.3 Functional Requirements

REQ-9.1	The system shall route a query to the escalation queue when intent confidence is below the configured threshold or the mapped workflow fails.
REQ-9.2	The system shall notify available human agents of a new escalated conversation.
REQ-9.3	The system shall allow a human agent to accept and respond to an escalated conversation.
REQ-9.4	The system shall deliver the human agent's response to the user through the conversation interface.
REQ-9.5	The system shall notify the user when no human agent is immediately available.

5.10 Conversation History and Notifications

5.10.1 Description and Priority

This feature allows users to view their past conversations with role assistants and receive notifications regarding escalation and resolution events.

Priority: Medium Benefit: 7, Penalty: 5, Cost: 4, Risk: 2

5.10.2 Stimulus/Response Sequences

- User opens "MyConversations" → System displays a list of the user's past conversations with relevant role assistants.
- User selects a specific conversation → System displays the full transcript and outcome.
- A conversation is escalated or resolved → System generates a notification for the user.
- User opens a notification → System displays the related conversation and its current status.

5.10.3 Functional Requirements

REQ-10.1	The system shall store the transcript and outcome of every conversation between a user and a role assistant.
REQ-10.2	The system shall allow a user to view a list of their past conversations.
REQ-10.3	The system shall display the full transcript and outcome when a user selects a specific conversation.

REQ-10.4	The system shall generate a notification for the user when a conversation is escalated or resolved.
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5.11 Admin Dashboard and Monitoring

5.11.1 Description and Priority

This feature provides administrators with a central interface to monitor role assistant usage, escalation activity, and overall system health.

Priority: Medium Benefit: 7, Penalty: 5, Cost: 5, Risk: 3

5.11.2 Stimulus/Response Sequences

- Administrator login → System authenticates the administrator and displays the admin dashboard.
- Administrator opens the dashboard → System displays summary statistics such as active role assistants, conversation volume, and escalation rate.
- Administrator selects a role assistant → System displays usage details specific to that role.
- An unauthorized user attempts to access the dashboard → System denies access and displays an appropriate error.

5.11.3 Functional Requirements

REQ-11.1	The system shall provide an administrator dashboard summarizing role assistant usage and status.
REQ-11.2	The system shall display conversation volume and escalation rate statistics on the dashboard.
REQ-11.3	The system shall allow an administrator to drill down into usage details for a specific role assistant.
REQ-11.4	The system shall restrict access to the administrator dashboard to authenticated and authorized administrator accounts.

5.12 Reports and Analytics

5.12.1 Description and Priority

This feature allows administrators to generate and view reports on intents handled, escalation frequency, and role-wise usage, supporting continuous improvement of role configurations.

Priority: Low Benefit: 6, Penalty: 3, Cost: 6, Risk: 4

5.12.2 Stimulus/Response Sequences

- Administrator opens "Reports and Analytics" → System retrieves available conversation data and displays relevant reports.
- Administrator applies a filter (e.g., by role, date range, or intent) → System processes the data and displays the filtered results.
- Administrator requests updated analytics → System refreshes the report using the latest available data.

5.12.3 Functional Requirements

REQ-12.1	The system shall allow authorized administrators to generate reports based on available conversation and escalation data.
REQ-12.2	The system shall provide analytics on intent frequency, escalation rate, and role-wise usage.
REQ-12.3	The system shall allow administrators to filter reports by role, date range, or intent.
REQ-12.4	The system shall present generated reports and analytics in a clear and understandable format.

5.13 Role-Based Access Control

5.13.1 Description and Priority

This feature ensures that users, administrators, and human agents can access only the system functions and information authorized for their respective roles.

Priority: High Benefit: 9, Penalty: 8, Cost: 4, Risk: 5

5.13.2 Stimulus/Response Sequences

- A user, administrator, or human agent logs in → System authenticates the account and identifies its authorized role.
- A user accesses the system → System provides access only to the functions permitted for the user role (role selection, conversation, history, escalation).
- An administrator accesses the system → System provides access to authorized administrative functions (role configuration, integration management, dashboard, reports).
- A user attempts to access an administrator-only or agent-only function → System denies access and displays an appropriate authorization error.

5.13.3 Functional Requirements

REQ-13.1	The system shall support role-based access control for users, administrators, and human agents.
REQ-13.2	The system shall restrict users to the functions and data authorized for their role.
REQ-13.3	The system shall provide administrators with access only to authorized administrative functions.
REQ-13.4	The system shall provide human agents with access only to the escalation queue and related conversation data.
REQ-13.5	The system shall enforce authorization checks before allowing access to protected system functions and data.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- **NFR-P1:** The system shall convert captured voice input to text within a reasonable response time under the typical prototype workload.
- **NFR-P2:** The system shall recognize the intent of a query and return a response within a reasonable response time under the typical dataset size used during testing and demonstration.
- **NFR-P3:** The administrator dashboard shall load usage statistics within a reasonable response time under the expected prototype workload.
- **NFR-P4:** Reports and analytics operations shall return results within a reasonable response time under the expected prototype workload.
- **NFR-P5:** Escalated conversations shall be delivered to the notification queue for human agents without unnecessary delay.

6.2 Safety Requirements

The Role-Specific AI Voice Assistant Framework is a web-based information and workflow-automation system and does not directly control physical equipment or safety-critical processes.

- **NFR-S1:** The system shall prevent accidental loss or modification of role-profile, workflow, and conversation data by validating important configuration operations before they are processed.
- **NFR-S2:** The system shall preserve conversation and escalation information required for tracking, auditing, and analytics.
- **NFR-S3:** The system shall display appropriate error messages when invalid or incomplete configuration or query information is entered so that users and administrators can correct it before submission.
- **NFR-S4:** The system shall require confirmation for important configuration operations, such as deleting a role profile or disabling an integration, where accidental actions could affect assistant availability.
- **NFR-S5:** No formal safety certification is required for the current academic prototype because the system does not operate or control safety-critical physical systems.

6.3 Security Requirements

- **NFR-SEC1:** The system shall require users, administrators, and human agents to authenticate using valid account credentials before accessing protected system functions.
- **NFR-SEC2:** The system shall protect user authentication credentials and shall not store passwords in plain text.
- **NFR-SEC3:** The system shall use secure communication mechanisms such as HTTPS during deployment to protect user credentials, voice data, and conversation content transmitted between the client and server.
- **NFR-SEC4:** The system shall enforce role-based access control at the application level so that users cannot access administrator-only or agent-only functions.
- **NFR-SEC5:** The system shall restrict access to personal information and conversation content to authorized users, administrators, and human agents according to their roles.
- **NFR-SEC6:** The system shall prevent unauthenticated users from accessing protected system functions.
- **NFR-SEC7:** The system shall maintain records of relevant administrative activities, including role profile changes, integration configuration changes, and escalation handling, to support accountability.
- **NFR-SEC8:** The system shall follow appropriate security and privacy practices for protecting user and conversation information within the scope of the academic prototype.
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6.4 Software Quality Attributes

- **Usability:** The system shall provide a simple, clear, and consistent voice/chat interface that allows users to converse with a role assistant without requiring specialized training.
- **Availability:** The system shall be available and functional during scheduled testing, demonstration, and evaluation periods.
- **Reliability:** The system shall handle invalid or unrecognized inputs and unexpected application conditions gracefully without crashing or causing loss of conversation data.
- **Scalability:** The system shall use a design that allows additional role profiles and integrations to be added without redesigning the core conversational engine.
- **Maintainability:** The system shall use a structured codebase with clearly separated speech processing, intent recognition, workflow execution, and data-access components so that individual features can be modified without unnecessarily affecting unrelated functionality.
- **Data Integrity:** The system shall maintain accurate and consistent role-profile, conversation, escalation, and integration data throughout the conversation lifecycle.
- **Accuracy:** The system shall provide sufficiently accurate intent recognition results for the intents configured during system testing.
- **Fault Tolerance:** The system shall handle speech-processing and integration errors gracefully and shall provide appropriate fallback responses without terminating the entire application.
- **Compatibility:** The web application shall function correctly on commonly used modern web browsers that support microphone access, including Google Chrome, Mozilla Firefox, and Microsoft Edge.
- **Auditability:** The system shall maintain relevant records of role-profile changes, integration changes, and escalation handling to support traceability and accountability.
- **Privacy:** The system shall limit access to personal information and conversation content to authorized users, administrators, and human agents and shall avoid unnecessary exposure of such information.
- **Accessibility:** The system shall provide readable text, clear labels, understandable navigation, and voice interaction to support users with different levels of digital familiarity.
- **Relative Priority:** For this system, accurate intent recognition and reliable escalation to a human agent are prioritized over advanced customization, since users must be able to get a correct answer or reach a human when the assistant cannot help.

6.5 Business Rules

- **BR1:** Only registered and authenticated users shall be permitted to converse with a role assistant.
- **BR2:** Users shall be permitted to view and access only their own conversation history.
- **BR3:** Only authorized administrators shall be permitted to create, edit, or deactivate role profiles.
- **BR4:** Only authorized administrators shall be permitted to configure or manage integrations with internal organizational systems.
- **BR5:** A query shall be escalated to a human agent whenever intent confidence falls below the configured threshold or the mapped workflow cannot be completed.
- **BR6:** Escalated conversations shall be made available only to authorized human agents for handling.
- **BR7:** Only authorized administrators shall be permitted to access the administrative dashboard and reports and analytics.
- **BR8:** The system shall maintain each conversation as an individual record and shall preserve its relevant information throughout the conversation lifecycle.
- **BR9:** Conversation and user information shall be accessible only according to the user's authorized role.

Domain Requirements

- **DR1:** Each role assistant shall operate using a distinct set of intents and workflow mappings specific to its organizational role.
- **DR2:** Each conversation shall maintain information about its current status (in progress, resolved, escalated) so that its progress can be tracked.
- **DR3:** A query that cannot be resolved by the assistant shall be capable of being escalated to a human agent.
- **DR4:** Workflow execution shall be capable of invoking configured internal-tool integrations to complete a user request.
- **DR5:** Personal information exchanged during a conversation shall be protected and shall not be unnecessarily exposed to unauthorized users.

7. Other Requirements

- **Legal/Compliance Requirement:** The system shall clearly state that it is an academic prototype developed as part of the Jackfruit mini-project and is not an official production assistant of any organization.
- **Scope Exclusion Requirement:** The system shall clearly indicate that requests requiring legal, medical, or emergency response are outside the scope of the assistant and should be directed to the appropriate authority or emergency service.
- **Data Retention Requirement:** The system shall retain conversation records and related information for as long as required for tracking, escalation handling, reporting, and system evaluation.
- **Database Requirement:** The system shall use a MySQL database to store user information, role profiles, intents, workflow mappings, integration configuration, conversation history, escalation records, and analytics data.
- **Data Backup Requirement:** The system should support appropriate backup and recovery procedures for preserving role-profile, integration, and conversation data.
- **Reuse Requirement:** Common components such as the conversation interface, authentication forms, and notification components should be designed for reuse across all role assistants to improve consistency and reduce development effort.
- **Future Enhancement Requirement:** Possible future enhancements may include support for additional languages, sentiment-aware responses, proactive notifications, and integration with real production HRMS, ITSM, and CRM systems. These enhancements are outside the scope of the current project.

Appendix A: Glossary

Term	Definition
Role Assistant	A configured instance of the AI voice assistant tailored to a specific organizational role (e.g., HR, IT Support, Admissions).
Role Profile	A configuration record defining a role assistant's name, supported intents, sample utterances, and workflow mappings.
Intent	The purpose or goal behind a user's query, recognized by the NLU engine (e.g., "check leave balance").
Entity	A specific piece of information extracted from a user query that is required to execute a workflow (e.g., a date or ticket category).
Workflow	A predefined sequence of actions, including internal-tool integration calls, executed in response to a recognized intent.
Escalation	The process of routing a query to a human agent when the assistant cannot resolve it with sufficient confidence.
NLU	Natural Language Understanding – the component responsible for intent recognition and entity extraction.
STT	Speech-to-Text – the component that converts spoken audio into text.
TTS	Text-to-Speech – the component that converts text responses into spoken audio.
Integration Connector	A configurable module that allows the workflow automation engine to call an internal organizational system's API.

Role-Based Control (RBAC)	Access	A security mechanism that restricts system functions and information according to the user's authorized role.
Human Agent		Support staff who receive and resolve conversations escalated by the assistant.
Flask/FastAPI		The Python-based web application framework used to develop the backend of the system.
MySQL		The relational database management system used to store and manage system data.

Appendix B: Field Layouts

Sample field layout for the Role Profile / Intent registration data, to be maintained as a companion spreadsheet containing field layouts, properties/attributes, and report requirements.

Field	Length	Data Type	Description	Mandatory
Role Name	50	String	Name of the role assistant (e.g., HR Assistant)	Y
Intent Name	50	String	Unique name of the intent within the role	Y
Sample Utterances	255	Text	Comma-separated example phrases for the intent	Y
Workflow ID	16	Alphanumeric	Identifier of the workflow mapped to the intent	Y
Integration Endpoint	255	String	URL/endpoint of the internal system integration, if applicable	N
Confidence Threshold	5	Numeric	Minimum confidence score required to avoid escalation	Y
Conversation ID	16	Alphanumeric	Unique identifier of a conversation record	Y
Escalation Status	20	Alphanumeric	Status of escalation (None/Pending/Resolved)	Y

Sample Report Requirements

Usage Report	Escalation Report
Role Assistant Name	Conversation ID
Conversation Count	Role Assistant Name
Top Intents Handled	Escalation Reason
Average Confidence Score	Assigned Human Agent
Escalation Rate	Escalation Timestamp
Date Range	Resolution Timestamp

Appendix C: Requirement Traceability Matrix

This matrix will be populated progressively during Phases 2 and 3 of the project, linking each requirement to its architecture reference, design reference, code file, and test case identifiers.

SI .N o	Req. ID	Brief Description	Arch. Ref.	Design Ref.	Code Ref.	Test Case ID	Sys. Test ID
1	REQ-1. 1	User registration	<TBD>	<TBD>	<TBD>	<TBD >	<TBD>
2	REQ-3. 1	Voice query capture	<TBD>	<TBD>	<TBD>	<TBD >	<TBD>
3	REQ-4. 1	Intent recognition	<TBD>	<TBD>	<TBD>	<TBD >	<TBD>
4	REQ-5. 2	Workflow execution	<TBD>	<TBD>	<TBD>	<TBD >	<TBD>
5	REQ-9. 1	Escalation to human agent	<TBD>	<TBD>	<TBD>	<TBD >	<TBD>